FALL 2026 · READER DECK

Trajectory Extraction, Cross-Embodiment Retargeting & TeleGuide

A research reference pack covering prior work, Panda/UR5e replay, pose uncertainty, budgeted correction, and reproducible experiments.

Reference material for senior CS students

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Purpose and use of this reference pack

READING OBJECTIVES

01

Assess existing evidence

Run clean-state physics on a CPU. Examine the schema, model receipt, and failure logs before introducing vision.

02

Define a research question

Panda/UR5e transfer and object-relative representations have precedents. The candidate question concerns correction-time allocation.

03

Design a controlled study

Establish that noisy pose inputs affect the plan and task outcome. Distinguish assumed oracle costs from measured human time.

Reference material for students and supervisors, who determine project scope and assessment.

Reader guide and sources

Shared technical core and distinct objectives

CURRENT RESEARCH PRIORITY

IL (Invested Lamda)

- Reusable pipelines and benchmarks with provenance.
- Measured capabilities and execution failures.
- Research prototypes before product expansion.

RELATED RESEARCH AND EDUCATIONAL OUTPUTS

CS support and future research

- CS: reference code, reading guides, and evidence.
- Future study: research questions and controls.
- Students and supervisors define capstone scope.

Share code, schemas, and evidence while preserving decision ownership and the scope of each conclusion.

Reader guide and sources

Pipeline contracts and levels of evidence

Each stage requires its own inputs and metrics. Object pose alone is insufficient to reconstruct a human manipulation task.

Video / robot log

RGB or RGB-D, calibration, timestamps

→

Pose and events

Object/TCP poses, grasp, phases, scale

→

Retargeting

Target IK, controller, contact, constraints

→

Correction

Segment selection, editing, replay, budget

Perception

Pose error, coverage, quality ranking

Kinematics / physics

IK residuals, failure codes, contact

Research endpoint

Task success and human supervision time

Reader guide and sources

SECTION 01

Simulation reference

Clean-state replay separates frame, grasp, and controller failures from video-estimation errors.

Input assumptions and evaluation objectives

- Human video or robot logs with TCP/gripper state?
- Available RGB-D, calibration, and metric CAD?
- Rigid-object tasks or richer contact interactions?
- Source of target object and goal poses?

- Replay, learned-policy, or assisted-success endpoint?
- Editing time or total supervision budget?
- Independent evaluation episodes and layouts?

These questions guide the choice of prior work and experiments without prescribing a student project.

Reader guide and sources

Implemented MuJoCo reference

Implemented and tested

- Panda source and UR5e + Robotiq 2F-85 target.
- 6D IK, position servos, and gravity compensation.
- Contact physics, a free-joint cube, and trial logs.

Scope of current evidence

- One cube and one source demonstration.
- Clean simulator state without video estimation.
- 20 tests passed, with no skipped tests.

CPU reference

Clean pick-and-place

50 development and 100 held-out target layouts.
Pinned Menagerie models with a hash receipt.

Reader guide and sources

Isaac and larger vision models

Potential benefits to establish

- Rendering or sensing beyond the CPU baseline.
- Parallel environments or dexterous tasks.
- Mimic/learning for a defined policy endpoint.
- Reproducible controllers and input contracts.

Resource requirements and fallback

- Pin compatible software, models, and drivers.
- Measure memory and runtime on a small input.
- Retain CPU replay for failure diagnosis.

Neither reference currently requires GPU rental. New vision or cloud workflows require a separate feasibility pilot.

Reader guide and sources

Evidence-based expansion of the reference

Stage	Experimental basis	Required evidence	Current status
P0 · Clean	MuJoCo on CPU	Grasp, transport, release, and logs	Measured: cube / 1 demo
P1 · Noisy	State noise + calibrated video	Noisy inputs affect plan and success	Integration proposed
P2 · Learning	ACT / Diffusion / Isaac as needed	Defined endpoint and usable data	Optional future stage

P0 is available. P1 requires physics integration. P2 depends on a learning or scale endpoint.

Reader guide and sources

SECTION 02

Literature and research gap

Individual components have precedents. A contribution requires a specific protocol and empirical evidence.

Replay and imitation learning

Demonstration
Observations, actions, timestamps, labels

→

Behavior cloning
Learn actions from recorded observations

→

Policy rollout
The policy encounters a shifted state distribution

→

Correction data
Interventions with context and outcomes

ACT

Action chunks are an option when the task requires a policy learned from demonstrations.

Diffusion Policy

A model of action distributions. Training requires a suitable schema and a defined endpoint.

MimicGen

A precedent for object-relative demonstration generation. Examine sensitivity to pose noise.

Reader guide and sources

Evidence cards for primary papers

1

Input

RGB, depth, or robot state? What establishes scale and calibration?

2

Output

Object pose, TCP trajectory, joint action, or learned policy?

3

Representation

How does the representation preserve contact or task intent?

4

Human cost

Who reviews, selects, and edits? Is the budget actions, samples, or seconds?

5

Evaluation

Trial units, counts, seeds, holdouts, success criteria, and failures?

6

Limitations

What is explicit, inferred, or still unverified?

Reader guide and sources

Object-relative transfer and morphology

Panda and UR5e differ

- Arm degrees of freedom: Panda 7, UR5e 6.
- Different link lengths, limits, and grippers.
- Joint copying does not preserve contact.

Precedents and target constraints

- MimicGen includes Panda/UR5e transfer.
- The target solves IK using its TCP convention.
- Track reachability, joint limits, and singularity.
- Evaluate contact success in physics or trials.

Current retargeting uses initial-object and goal anchors. Editing later object poses need not change the plan.

Reader guide and sources

Automatic recovery: PSI and RIGVid

CLOSEST AUTOMATIC COMPARATORS

Filtering and backtracking

PSI validates trajectories and grasps in simulation. RIGVid tracks pose and backtracks after deviations. Reproduction requires their methods and input assumptions.

Filtering | Recovery | Pose inputs

BASELINE INTERPRETATION

- PSI differs on all/any failed-grasp rejection.
- Its 2 cm/0.2 rad threshold belongs to ICP.
- RIGVid uses a depth anchor and feedback.
- An internal filter requires a separate label.

Automatic recovery and human pose correction differ in information, feedback, and cost. Preserve these distinctions.

Reader guide and sources

Reliability and budgeted human intervention

STRUCTURAL PRECEDENTS

ThriftyDAgger and VR-DAgger

Robot-triggered intervention is established. VR-DAgger selects a timestep using action uncertainty, then teleoperates the remainder. Its budget counts correction samples.

LIMITS OF TRANSFER

- Meta Pose: rotation failures in individual images.
- SPARC: bounding-box annotation reliability.
- Neither establishes the utility of object-pose segment selection for retargeted success.

Candidate gap: reliability-guided pose correction with a seconds budget, random selection, and automatic comparators.

Reader guide and sources

Reading sequence and design decisions

Design question	Priority reading	Evidence to inspect
Representation	MimicGen	Object anchors, Panda/UR5e, pose-noise Appendix U
Automatic recovery	PSI · RIGVid	Rejection/grasp rules, depth anchors, backtracking
Human budget	ThriftyDAgger · VR-DAgger	Actions/samples/seconds, review cost, controls
Reliability	Meta Pose · SPARC · ProxyPose	Failure labels, scale, and task sensitivity
Learning (optional)	ACT · Diffusion Policy	Observations/actions, loaders, policy endpoint

Inspect experiments, appendices, and code. Separate explicit claims, interpretations, and unresolved evidence.

Reader guide and sources

SECTION 03

Experiments and reference results

Noisy inputs must affect the plan and task outcome before correction-allocation strategies can be evaluated.

Candidate research question

For a correction budget of B seconds, does reliability-guided segment selection improve task success over random selection and automatic recovery?

Correction target

Video-derived object poses. Separate global biases from local failures.

Controlled comparison

Shared candidates, edit primitives, controller, and budget accounting.

Primary endpoint

Task success after retargeting. Include a GT-assisted oracle comparison.

The question remains untested. A negative result is informative after protocol and input sensitivity are established.

Reader guide and sources

Pose inputs and data contracts

1 · Input
Calibration, timestamps, scale provenance

→

2 · Estimate
SE(3), validity mask, diagnostics

→

3 · Events
TCP, gripper, phases, and goal

→

4 · Retarget
Requested plan and executed states

→

5 · Evaluate
Task criteria, failures, and cost

Frame conventions

T_A_B maps B to A. Use metres, seconds, and wxyz quaternions. Retain missing frames.

Ground-truth isolation

GT enters the evaluator/oracle only. Estimated poses must reach the planner without a clean-state shortcut.

Training export

LeRobot/HDF5 requires an adapter and loader roundtrip. Physics JSON alone is not an ACT/Diffusion dataset.

Reader guide and sources

Clean-state physics: held-out results

Method	Success / N	Wilson 95% CI	Failures
World-frame	0 / 100	0.00–3.70%	94 grasp, 2 drop, 4 place
Object-relative	89 / 100	81.37–93.75%	11 IK failures
Object-relative trimmed	100 / 100	96.30–100.00%	None in this sample
Target scripted reference	100 / 100	96.30–100.00%	None in this sample

One cube, one source demonstration, 100 target layouts at seed 7. Clean state without video or correction.

Reader guide and sources

Module contracts and integration

Module	Principal contract	Implementation status
Physics replay	TCP/object/goal, grasp command, task criteria	Clean demonstration and benchmark
Video / marker	Pixels, calibration, estimates, validity mask	Synthetic/PnP pilot; translation only
Retarget input	Initial object anchor + TCP/goal trajectory	65 edited pose frames leave plan unchanged
Correction / cost	Edited artifacts, candidates, time receipts	Assumed oracle cost; terminal editor
Bridge / export	Consequential pose inputs, physics, labels	End-to-end integration remains absent

The missing bridge: edited inputs must alter the plan before downstream success can be assessed.

Reader guide and sources

SECTION 04

TeleGuide and correction budgets

An intervention must produce a replayable correction artifact, time records, and an observable outcome.

Review, correction, and reusable data

REVIEW

Inspect context and intent

Pose axes, diagnostics, source frames, and candidate segments. Record review and selection time.

CORRECT

Use a shared edit primitive

Offsets, waypoints, or phases under a defined contract. Save a new artifact and preserve the original.

REPLAY

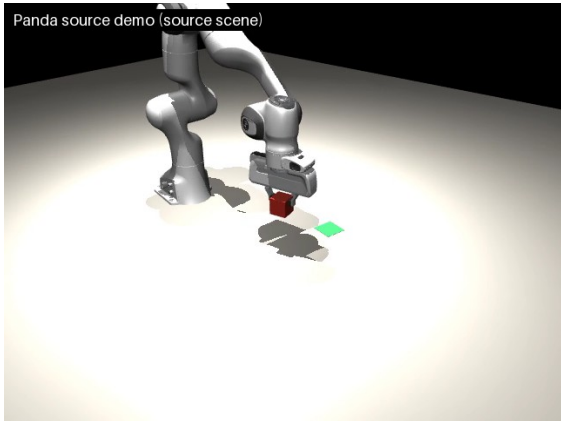
Outcome and provenance

Retarget again and retain failures. Record authority labels when exporting training data.

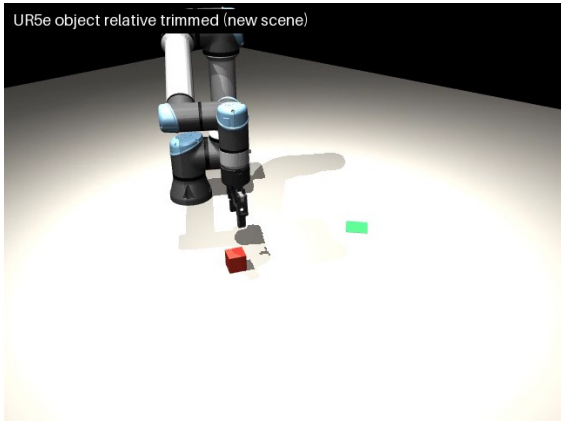
The terminal editor measures session elapsed time. Active editing, review, and waiting remain unseparated.

Reader guide and sources

Manipulation demonstration in MuJoCo



Panda source · simulator state



UR5e trimmed · contact physics

Engineering evidence

- Panda source, UR5e target.
- Contact physics.
- Clean simulator state.
- Predicate-based success.

Not yet evaluated

Video/correction is not integrated with physics. No human study or real-robot evaluation.

Reader guide and sources

A protocol with matched correction budgets

Candidate pool
Segments, lengths, costs, validity

→

Conditions
Automatic, random, ranked, oracle

→

Correction
Shared primitive and observable information

→

Retarget/replay
Shared layouts, controller, success criteria

Reported measurements

Task success, rejects, review/edit/wait/total time, pose error, and failure taxonomy.

Human-cost comparison

Active-time reduction requires measured full teleoperation at matched task quality.

Reader guide and sources

Resources, risks, and contingency measures

Risk	Contingency / diagnostic	Retained evidence	Status
Poor poses / occlusion	Validity mask, filtering, manual edit	Original/estimated/corrected data + reasons	Proposed physics integration
Clean replay failure	Frame/TCP, grasp, controller, IK	Failure logs and reduced task scope	Reference available for diagnosis
Idle GPU / cost	Heartbeat, checkpoint, then stop	Runtime/cost ledger and receipt	Operational measures proposed
Data / checkpoints	Hashed local cache; selective sync	Manifest, version, retention policy	Storage procedure proposed

The CPU reference is available. Cloud idle-timeout and caching procedures remain proposed.

Reader guide and sources

An experimental sequence

STEP A

Pose sensitivity

- Consequential clean/corrupt inputs.
- Anchor, frame, and scale assumptions.
- Failure logs and physics replay.

STEP B

Correction protocol

- Edited artifacts alter the plan.
- Random/ranked use matched budgets.
- Oracle pilot, then measured human time.

STEP C

Calibrated video

- Markers or RGB-D with a reference.
- Occlusion and scale diagnostics.
- Markerless estimation when feasible.

Clean replay failure: examine frames, TCP, grasp, controller, and IK before expanding scope.

Reader guide and sources

Evidence checklist for the reference pack

Technical resources

- Clear CPU setup and run commands.
- Schema and model/config/code hashes.
- Example episodes and per-trial failures.
- Clean replay across two embodiments.
- Videos and results with explicit scope.

Research resources

- Closest works and verified corrections.
- Research questions, controls, and budgets.
- Trial definitions and uncertainty.
- Negative results and limitations.
- Proposed, tested, and evaluated states.

This checklist supports reproducibility. It does not replace the institution’s capstone assessment criteria.

Reader guide and sources

Repository execution paths

Physics reference

- Python 3.12 in a separate environment.
- pip install -r physics_replay/requirements.txt
- python -m physics_replay fetch-models
- python -m physics_replay demo
- python -m pytest -q physics_replay/tests

Marker / correction prototype

- Separate environment: requirements-research.txt.
- python tools/research-prototype.py fetch-models
- python tools/research-prototype.py demo
- A new output directory for each run.
- See docs/marker-prototype.md.

Public starter repository. Separate environments and endpoints. End-to-end integration remains absent.

Reader guide and sources

Resources: repository and documentation

Links provide code, notebooks, committed results, and reproducibility guidance.

Executed reference

Public code, notebook, and reference results. Run manifests retain model, configuration, and code hashes.

Physics README
physics_replay/README.md

Colab CPU notebook
Open the public CPU notebook

Held-out results
results/bench-heldout-seed7-n100

Example episode
panda_pick_place_10hz.json

Reading and provenance

Consult the README, proof status, and reports before citing a numerical result or research claim.

Reader guide and sources
docs/reader-guide.md

Marker / correction
docs/marker-prototype.md

Reading guide
docs/reading-guide.md

Resources: primary papers

Read the relevant versions, experiments, appendices, and code. Novelty remains conditional on source coverage.

Representation and recovery

Prioritize MimicGen, PSI, and RIGVid for the chosen inputs. Defaults do not establish paper trial counts.

MimicGen
https://arxiv.org/abs/2310.17596

PSI
https://arxiv.org/abs/2602.13197v2

RIGVid
https://arxiv.org/abs/2507.00990v3

ThriftyDAgger
https://arxiv.org/abs/2109.08273

ProxyPose
https://arxiv.org/abs/2607.06555v1

Reliability and intervention

Sample/action budgets differ from seconds. Pose ranking alone does not establish task utility.

VR-DAgger
https://arxiv.org/abs/2605.27114v2

Meta Pose
https://arxiv.org/abs/2608.03516v1

SPARC
https://arxiv.org/abs/2606.13497v1

ACT / ALOHA
https://arxiv.org/abs/2304.13705

Diffusion Policy
https://diffusion-policy.cs.columbia.edu/

Reader guide and sources